Mesoscope vs BWM: supervised decoding of task variables

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IBL mesoscope pilot analysis, v7, November 2, 2026

Abstract

We decoded stimulus side, choice, feedback and wheel speed from two-photon mesoscope imaging (151 sessions, 12 mice) and Brain Wide Map (BWM) Neuropixels recordings of dorsal cortex, with identical decoders and tests. Three results stand out. First, because each mesoscope session images many cortical regions at once, it can separate region from session effects, which single-probe recordings cannot. Which mouse and which session are recorded matter more for decoding than which region: scores from all neurons of a region vary more between sessions and animals (41–59% of the variance) than between regions (10–30%). Resampling whole animals confirms this for stimulus side, choice, feedback and trial-specific wheel speed; for wheel speed, where region matters most, it is uncertain. Session-to-session variation stays larger than region even within single mice. Within a mouse, regions do differ about as much as sessions for most variables, but which regions decode best differs between mice, so the region effect that holds across mice is small (Fig. 3). Region maps pooled across animals that each contribute few regions therefore mix region with animal- and session-specific differences. Second, wheel-speed neurons are rare in the imaged population, but not weaker: 3.5% of mesoscope neurons track wheel speed replicably against 22% of BWM units, a fraction that does not change with BWM recording depth. Yet the best 10 neurons of a mesoscope session predict held-out wheel speed as well as the best 10 BWM units (r 0.50 vs 0.47; Figs. 4, 6). Weak wheel decoding from random mesoscope neurons (R^2 0.05 vs 0.25 for 20 neurons) thus reflects sparseness, not imaging depth or the decoder, which passed every check (Fig. 7). Imaging and spike sorting sample different cells, so sparseness is a property of the imaged population. Third, most of any wheel-speed R^2 is the stereotyped speed profile after movement onset: predicting each time bin from its mean over training trials, with no neural data, already gives $R^2 \approx 0.45$. A trial-specific target, speed minus that mean, isolates what neurons add.

Introduction

The International Brain Laboratory (IBL) mapped neural correlates of its standardised visual decision-making task across the mouse brain with Neuropixels probes, in the Brain Wide Map (BWM) [1,2]. Two-photon mesoscope imaging gives a complementary view of dorsal cortex: many more neurons per session and many cortical regions recorded at once, but only superficial layers, slower calcium signals and a different sample of cells. How the two compare on the same task is not known.

Here we decode stimulus side, choice, feedback and wheel speed from IBL mesoscope sessions and from the BWM data, with identical trial criteria, decoders and single-neuron tests, and map the results across dorsal cortex. Because each mesoscope session covers several regions, it also lets us ask a question single-probe recordings cannot answer: do decoding scores differ more between regions or between sessions and animals?

Results

Overview. Both datasets are mapped in 1 mm hexagonal bins of dorsal cortex (coverage in Fig. S2). BWM is shown twice: units within 640 μm of the pial surface, the main comparison, and units at all depths. Table 1 gives the headline numbers. With 20 random neurons per bin, the mesoscope decodes stimulus side (0.54) and feedback (0.65 vs 0.63) as well as depth-limited BWM, but stays below it for choice (0.53 vs 0.60) and wheel speed (R^2 0.05 vs 0.25; Fig. 1). Restricting BWM to superficial units costs little (choice 0.60 vs 0.62, feedback 0.63 vs 0.69, wheel R^2 0.25 vs 0.31). With all neurons of a bin, about 20 times more in the mesoscope (median 458 vs 27), the mesoscope decodes stimulus side and feedback better than BWM, and its frontal feedback map stands out (Fig. 2).

Decoding from 20 random neurons per session and bin (mean over 10 draws) compares all three rows at the same population size; only sessions with at least 20 neurons in a bin enter it, and bins fade when fewer than four animals contribute (Fig. 1).

The bottom row of Fig. 2 decodes each BWM probe from all its isocortex units and places it at its brain entry point; per-probe medians match the per-bin ones.

	Mesoscope	BWM $\leq$640 μm	BWM all depths
Isocortex neurons (sessions)	1,071,534 (151)	2,929 (204)	13,002 (288)
Session $\times$ bin pairs with $\geq$20 neurons	1,763	54	216
Median neurons per session $\times$ bin ($\geq$20)	458	27	40
Fraction tuned: stimulus side / choice / feedback	0.147 / 0.107 / 0.254	0.075 / 0.090 / 0.233	0.069 / 0.108 / 0.294
Stimulus side, 20 neurons / all	0.542 / 0.620	0.538 / 0.546	0.541 / 0.551
Choice, 20 neurons / all	0.527 / 0.585	0.599 / 0.615	0.621 / 0.655
Feedback, 20 neurons / all	0.650 / 0.819	0.632 / 0.651	0.691 / 0.738
Wheel speed R^2, 20 neurons / all	0.046 / 0.215	0.246 / 0.286	0.306 / 0.385
Trial-specific wheel speed R^2, 20 neurons / all	0.012 / 0.076	0.052 / 0.065	0.073 / 0.110
Bin-means-only wheel R^2 (no neurons, median session)	0.47	0.44	0.44

Table 1: Decoding: balanced accuracy (chance 0.5) or R^2 (chance 0), mean over session $\times$ bin pairs. "All" uses every neuron of a session in the bin; mesoscope bins hold about 20 times more neurons, so only the 20-neuron columns compare like with like. Only 53 of the 204 sessions with superficial BWM units have a bin with at least 20 of them.

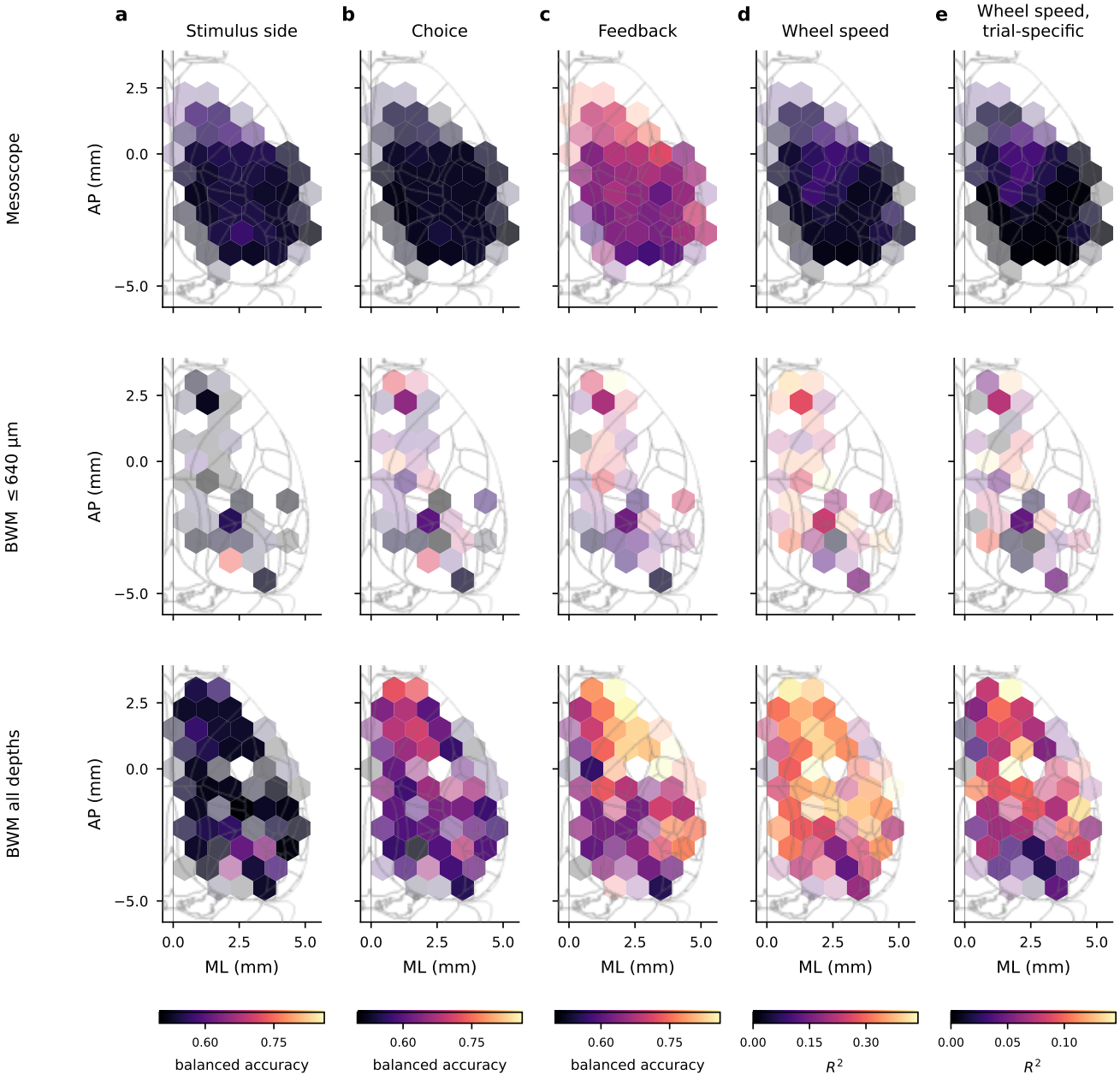


Figure 1: Decoding from 20 neurons. Rows: mesoscope; BWM units within 640 μm of the pial surface; BWM at all depths. Each bin (1 mm hexagon, hemispheres folded) shows the mean over sessions of decoders trained on 20 random neurons of one session in that bin (mean of 10 draws; sessions with $\geq$20 neurons there). Opacity: number of animals (full at $\geq$4). **a** Stimulus side, balanced accuracy (chance 0.5; colour scale shared by **a**–**c**). **b** Choice. **c** Feedback. **d** Wheel speed, cross-validated R^2. **e** Trial-specific wheel speed, R^2.

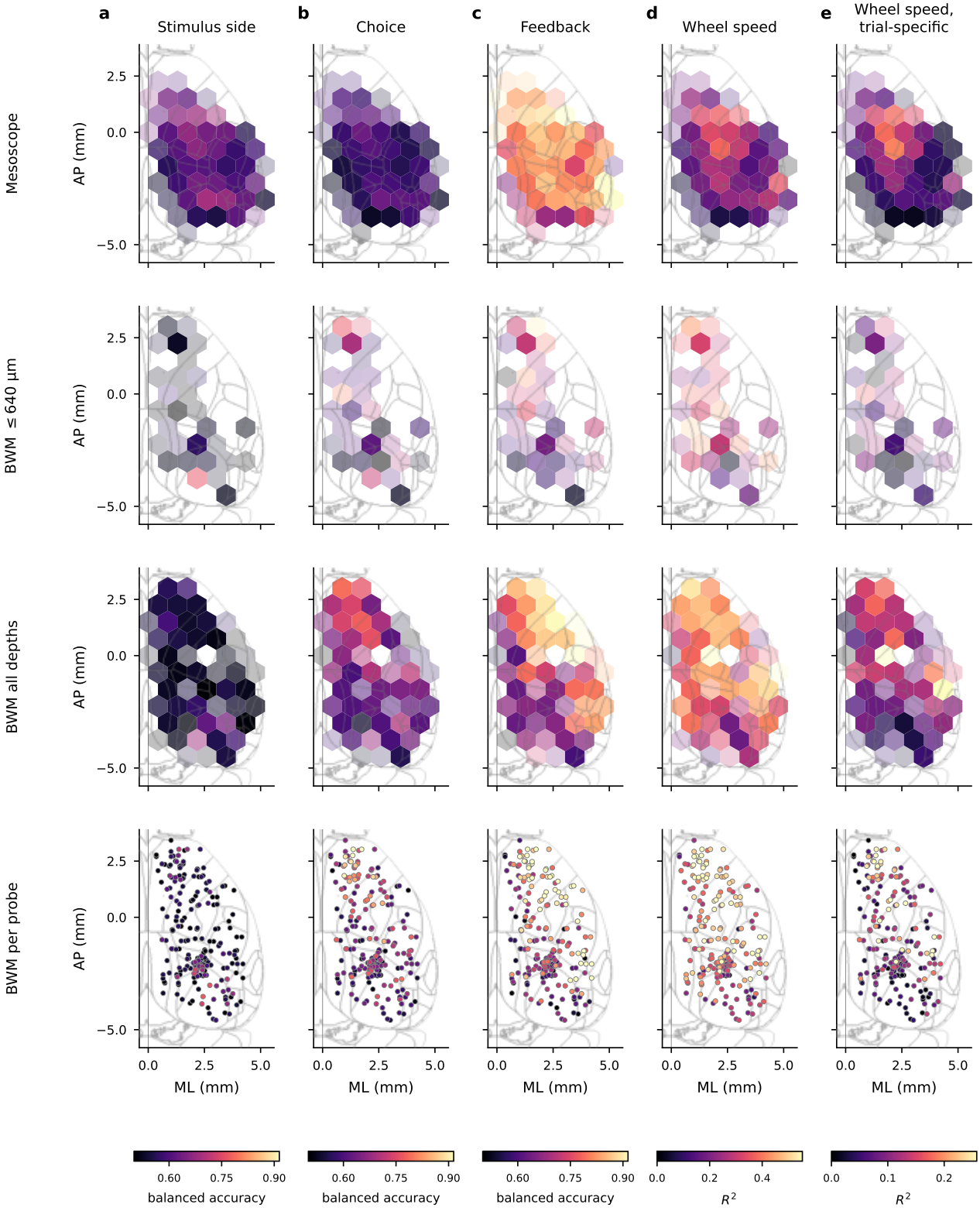


Figure 2: Decoding from all neurons. Rows: mesoscope; BWM units within 640 μm of the pial surface; BWM at all depths. Each bin (1 mm hexagon, hemispheres folded) shows the mean over sessions of decoders trained on all neurons of one session in that bin (sessions with $\geq$20 neurons there); mesoscope bins hold far more neurons than BWM bins. Opacity: number of animals (full at $\geq$4). Bottom row: BWM decoded per probe from all its isocortex units at all depths, whichever bins they fall in (209 probes with $\geq$20 units, median 40); each circle sits at the probe's brain entry point (histology-aligned Alyx trajectory, hemispheres folded), higher scores drawn on top where circles overlap; colour scales as in the rows above, so circles beyond the top of a scale saturate. **a** Stimulus side, balanced accuracy (chance 0.5; colour scale shared by **a**–**c**). **b** Choice. **c** Feedback. **d** Wheel speed, cross-validated R^2. **e** Trial-specific wheel speed, R^2.

Decoding varies more between sessions than between regions. Do all-neuron scores differ more between regions or between sessions? Scores are shown per Beryl region for the mesoscope (per session), BWM per probe (at its entry region) and the BWM decoding paper (per session and region), for the 15 regions sampled by both datasets. In both BWM versions frontal motor cortex stands out: MOp and MOs have the highest median scores for choice, feedback and wheel speed in the decoding paper. In the mesoscope, where every session samples several regions, region matters (mixed model, region share 10–30%, all $p < 0.001$, most for wheel speed), but sessions and animals together account for more (41–59%). With only 12 animals, we resampled whole animals (200 refits; error bars in Fig. 3b). Session plus animal exceeds region in 99–100% of resamples for stimulus side, choice and feedback and in 96% for trial-specific wheel speed. For wheel speed, where region matters most, it does so in only 78% and the intervals overlap, so there the comparison is uncertain. The model-free test agrees: within a region, scores vary more across sessions than across regions within a session, for every variable, and this holds when sessions are only compared within the same mouse (adjusted η^2 0.34–0.71 for session within mouse vs 0.13–0.36 for region; Fig. 3c). Fitting the model separately in each mouse (11 mice with at least 4 sessions) gives a more mixed picture (Fig. 3d): the session share exceeds the region share in 9 of 11 mice for feedback (median difference 0.38), but in only 3–6 of 11 for the other variables (medians -0.03 to 0.05). The reason is that a per-mouse region share also contains region differences specific to that mouse. Adding a region $\times$ mouse term to the pooled model shows these are substantial: they explain 11–17% of the variance (all $p < 0.001$), while the region effect shared across mice drops to 5–28% (stimulus side 15%, choice 7%, feedback 5%, wheel speed 28%, trial-specific wheel speed 23%). Within a mouse, regions therefore differ about as much as sessions for most variables, but which regions decode best varies between mice; only feedback is dominated by sessions even within mice. Example mice make this visible (Fig. 3e): in a session-dominated mouse (SP075) the score matrix is striped horizontally, with good sessions good in every region; in a region-dominated mouse (SP058) it is striped vertically, with some regions decoding well in most sessions. For ephys we do not have enough data to test how much region and session contribute to the variance: few BWM sessions record two or more of these regions, and those are usually adjacent along one probe. The two BWM rows of Fig. 3a therefore only show the scores, and panels b–e are mesoscope only.

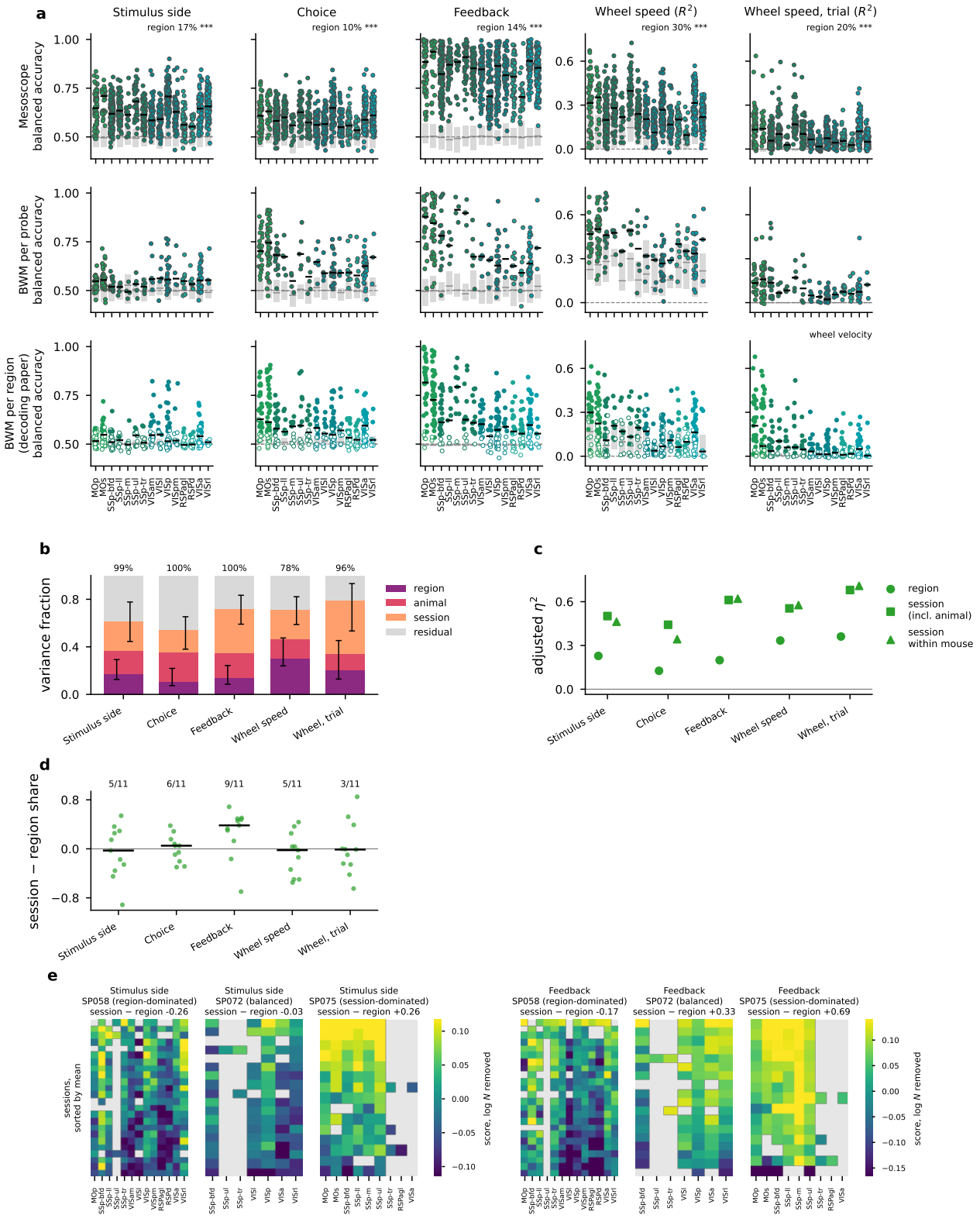


Figure 3: Variance of decoding across regions and sessions (all neurons). The 15 regions with ≥3 mesoscope sessions and BWM probes. Methods: *Region and session variance.* **a** Scores per region (columns ordered as in the Allen structure tree; Allen colours; bar: median; dashed: chance; grey band and line: 5–95th percentile and median of the baseline; for trial-specific wheel speed the shuffle baseline lies at $R^2 \approx 0$ and is hidden by the dashed line). Top: mesoscope, one dot per session and region from all its neurons there (149 sessions, 12 animals); baseline: shuffle control. Middle: BWM, one dot per probe from all its isocortex units, placed in the region where it enters the brain (198 probes, 93 animals; the majority region of its superficial units for 66% of probes); baseline: shuffle control. Bottom: BWM decoding paper, one dot per session and region from the region's units (346 pairs, 263 sessions, 116 animals); filled: $p < 0.05$ against pseudo-sessions (21–70% of dots); baseline: median of the pseudo-session null; wheel velocity in place of trial-specific wheel speed. Y axes shared within balanced accuracy and within R^2. Inset, top row: region share of variance from **b**. **b** Mesoscope variance fractions from a mixed model, score ∼ log N + (1|region) + (1|animal) + (1|session). Error bars: 95% intervals from resampling whole animals (200 refits), for the region share (left bar, at the top of the region block) and the session plus animal share (right bar, at the top of the session block); numbers above: fraction of resamples in which session plus animal exceeds region. Region: stimulus side 17%, choice 10%, feedback 14%, wheel speed 30%, wheel, trial 20%; session plus animal: stimulus side 45%, choice 44%, feedback 58%, wheel speed 41%, wheel, trial 59%. The region component is significant for every variable (likelihood-ratio test, $p < 0.001$). **c** The same question as **b**, do scores differ more between regions or between sessions, but without a statistical model, using only reshuffling: adjusted η^2 of region (circles; region labels shuffled within sessions), of session (squares; session labels shuffled within regions, across mice, so this includes differences between mice) and of session within mouse (triangles; shuffled within region and mouse, so differences between mice are removed, the model-free counterpart of **d**), 0 = chance. Filled: $p < 0.05$. **d** The same comparison within each mouse: mixed model score ∼ log N + (1|region) + (1|session) fitted separately per mouse with ≥4 sessions (11 mice); dots: session share minus region share per mouse; bar: median; numbers above: mice with a larger session share. **e** Example mice (with ≥10 sessions and >7 regions): the most region-dominated, the most balanced and the most session-dominated by **d**, averaged over stimulus side

Wheel-speed neurons are rare in the mesoscope, not weaker. Replicable wheel correlates are rare in the mesoscope: a median of 3.5% of neurons, against 22% of BWM units (Fig. 6). This BWM fraction is 20–25% at every 160 μm depth band, so depth does not explain the difference; choice and feedback tuning, in contrast, rise in deep layers, which the 640 μm cutoff mostly excludes (Fig. 4). The best 10 neurons of a whole mesoscope session predict held-out wheel speed as well as the best 10 BWM units (r 0.50 vs 0.47), but among 30 random neurons they fall far short (r 0.15 vs 0.37), so a random set of 20, as in one spatial bin, rarely contains a wheel neuron (Fig. 6).

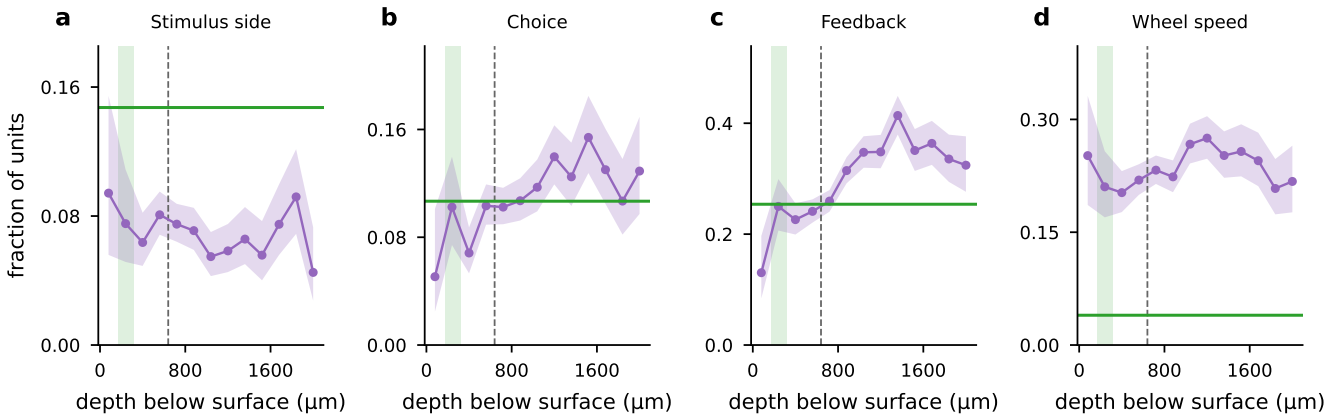


Figure 4: Depth sensitivity of the BWM comparison. Green line: mesoscope fraction over all neurons; green shading: mesoscope imaging depth (174–318 μm, 5–95 % of the histology-registered sessions); dashed line: the 640 μm cutoff of the depth-limited BWM set. Depth: Allen CCF top surface at the unit's ML-AP position minus its DV. **a** Fraction of BWM isocortex units tuned to stimulus side ($p < 0.05$, same test as the maps) per 160 μm depth band (purple; band: 95 % Wilson interval; bands with $\geq$30 units): 7.5% of BWM units within 640 μm, 6.7% deeper, mesoscope 14.7%. **b** Choice: 9.0% of BWM units within 640 μm, 11.4% deeper, mesoscope 10.7%. **c** Feedback: 23.3% of BWM units within 640 μm, 31.2% deeper, mesoscope 25.4%. **d** Fraction of units correlated with wheel speed ($r > 0.1$ over the wheel bins of both the even and the odd trials): 21.5% of BWM units within 640 μm, 23.2% deeper, mesoscope 4.0%.

Rasters show the same picture. In the two best mesoscope sessions, the mean of the 60 neurons most correlated with wheel speed, ranked on other trials, follows held-out wheel speed at $r = 0.58$ and 0.53, comparable to the best BWM sessions (0.29 and 0.63). But only 1–18% of mesoscope neurons reach $r > 0.1$ in these sessions, against 0–42% in BWM (Fig. 5).

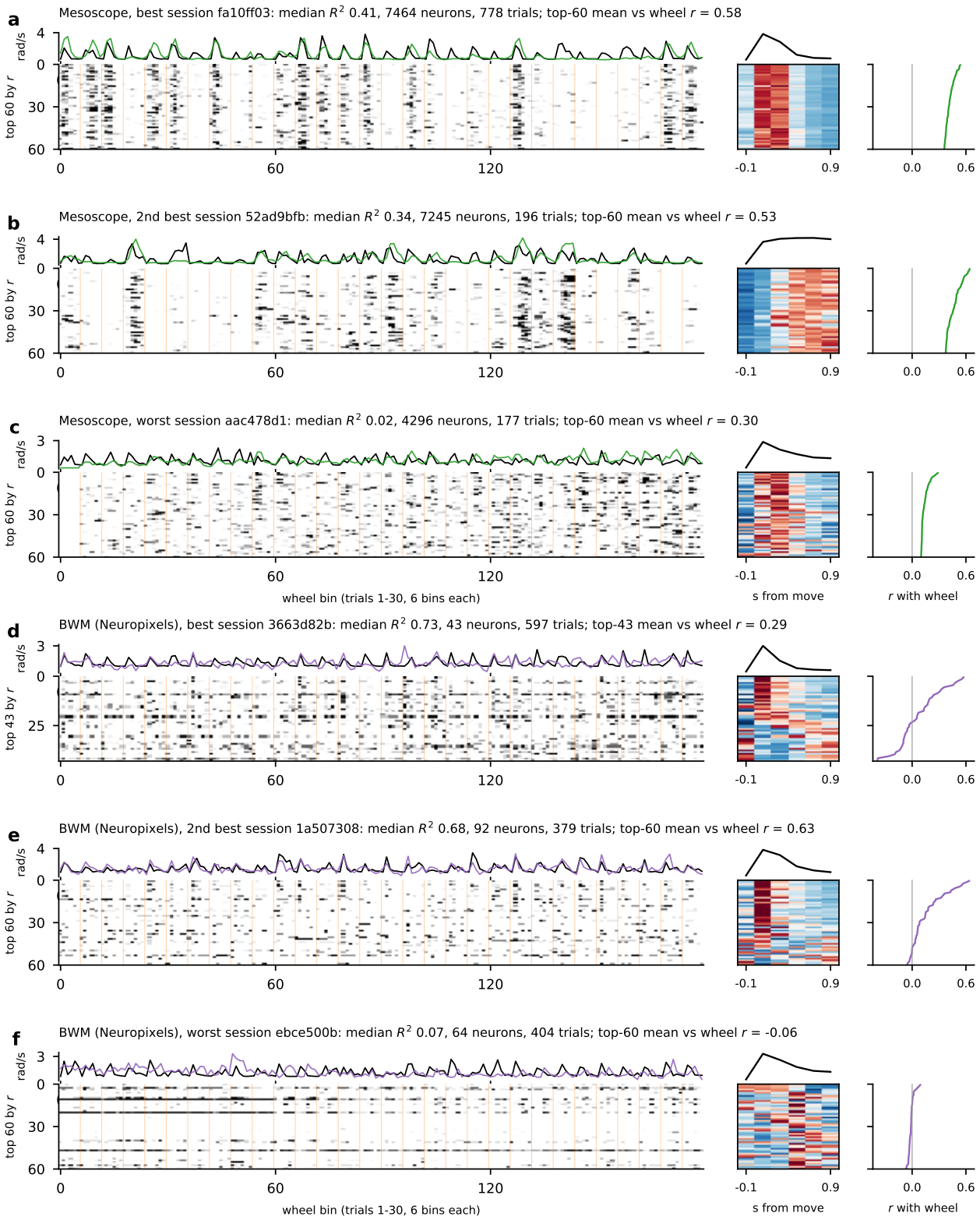


Figure 5: Neural correlates of wheel speed. Each row: wheel speed (top; rad/s) in the 6 wheel bins (0.2 s, -0.2 to 1 s around first movement) of the first 30 trials, concatenated (orange lines: trial boundaries; bins are not contiguous across trials; coloured: mean z-scored activity of the neurons below, rescaled), above the z-scored activity of the 60 neurons most positively correlated with wheel speed (sorted by r, highest on top). Neurons are ranked by r over all bins of the remaining trials, so the shown trials are held out from the selection. Middle: trial-averaged activity of the same 60 neurons per bin (z-scored per neuron, same order), with the mean wheel speed above. Right: their r with wheel speed over the remaining trials, same rows. Sessions at the best two and the worst of each modality by the session median R^2 over bins (all neurons per 0.4 mm bin, revised decoder with lagged features; sessions with $\geq$30 neurons). **a** Mesoscope session fa10ff03 (best of 149 sessions by the session median wheel-speed R^2): 95th percentile of r over neurons 0.20, maximum 0.54, 17% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = 0.58. **b** Mesoscope session 52ad9bfb (2nd best of 149 sessions by the session median wheel-speed R^2): 95th percentile of r over neurons 0.20, maximum 0.64, 18% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = 0.53. **c** Mesoscope session aac478d1 (worst of 149 sessions by the session median wheel-speed R^2): 95th percentile of r over neurons 0.07, maximum 0.29, 1% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = 0.30. **d** BWM (Neuropixels) session 3663d82b (best of 127 sessions by the session median wheel-speed R^2): 95th percentile of r over neurons 0.50, maximum 0.57, 42% of neurons with $r > 0.1$; mean of the top 43 neurons vs wheel speed in the shown trials r = 0.29. **e** BWM (Neuropixels) session 1a507308 (2nd best of 127 sessions by the session median wheel-speed R^2): 95th percentile of r over neurons 0.51, maximum 0.64, 30% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = 0.63. **f** BWM (Neuropixels) session ebce500b (worst of 127 sessions by the session median wheel-speed R^2): 95th percentile of r over neurons 0.05, maximum 0.09, 0% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = -0.06.

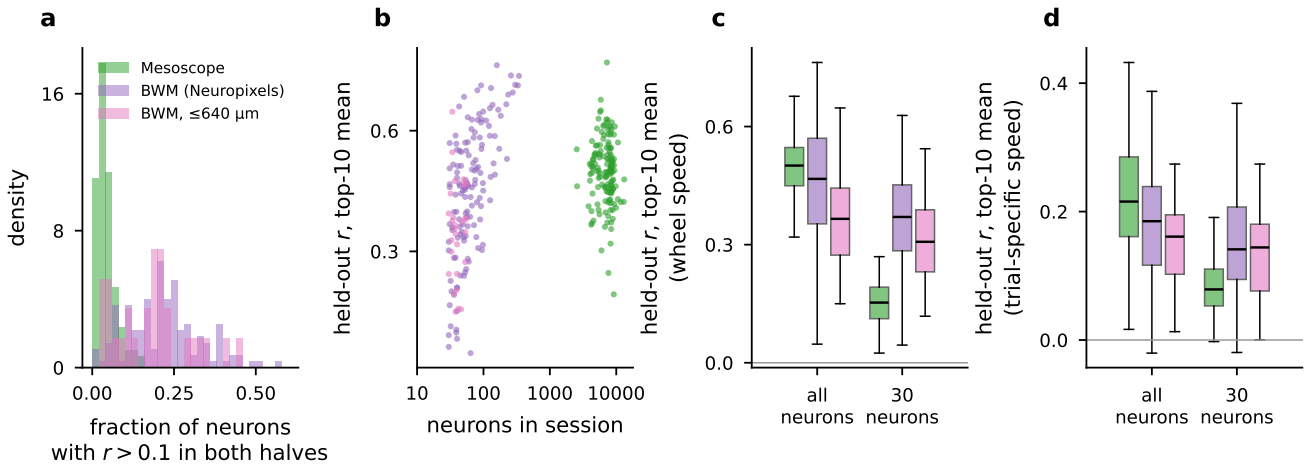


Figure 6: Wheel-speed correlates are rare in the mesoscope but present: among thousands of neurons per session the best 10 track wheel speed in held-out trials as well as the best BWM units, but a random set of 30 neurons, as in a small spatial bin, contains far fewer of them than a BWM set of the same size. Sessions with $\geq$30 neurons and $\geq$50 trials. **a** Fraction of neurons per session whose correlation with wheel speed (over all wheel bins) exceeds 0.1 in both the even and the odd trials: median 3.6% (mesoscope, 149 sessions), 21.7% (BWM isocortex, 137), 18.2% (BWM units within 640 μm of the surface, 29). **b** Correlation with wheel speed in the odd trials of the mean z-scored activity of the 10 neurons that correlate best on the even trials, against the number of neurons in the session (one dot per session). **c** The same with all neurons of a session (median mesoscope 0.50, BWM 0.47, p = 0.03) and with sessions subsampled to 30 random neurons (mean over 20 draws; 0.15 vs 0.37, $p < 10^{-32}$). Boxes: quartiles; whiskers: 1.5 inter-quartile ranges. **d** Same against trial-specific speed (wheel speed minus its per-bin mean over the even trials): 0.22 vs 0.19 (all neurons), 0.08 vs 0.14 (30 neurons).

Most of the wheel-speed R^2 is the stereotyped speed profile. The wheel decoder passed every check: it matches scikit-learn, it recovers an artificial neuron carrying wheel speed plus noise at the expected R^2, and the wheel speed per bin matches `brainbox` at $r \geq 0.9999$. Predicting each bin from its mean over training trials, with no neural data, already gives $R^2 \approx 0.45$ (Table 1): most of any wheel R^2 is the stereotyped speed profile after movement onset. Trial-specific wheel speed, speed minus that mean, isolates trial-to-trial differences and is decoded at R^2 0.01 (mesoscope) and 0.05 (depth-limited BWM) from 20 neurons. Mesoscope activity follows wheel speed by 0.15–0.25 s while BWM spikes lead it by 0.1 s, and using the previous, same and next bin as features raises wheel R^2 in both (mesoscope 0.015 to 0.026, BWM 0.144 to 0.218; 10 neurons, 0.4 mm bins; Fig. 7).

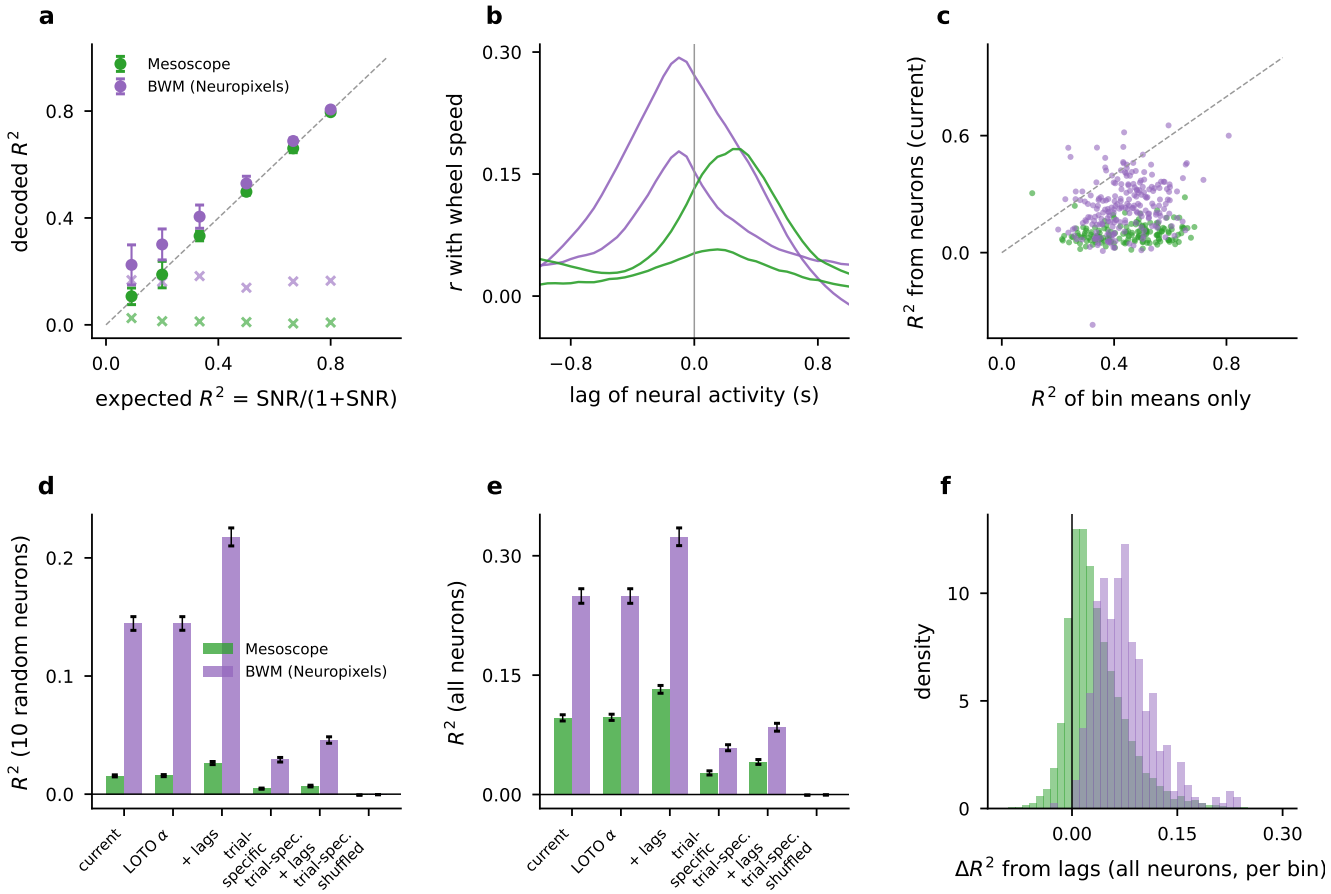


Figure 7: Checks of the wheel-speed decoding. The decoder and target are correct (after removing encoder glitches in 16 mesoscope sessions), mesoscope activity lags the wheel by about one 0.2 s bin more than spikes, and the R^2 is dominated by the stereotyped speed profile after movement onset rather than by trial-to-trial differences in speed. **a** Decoder check: a synthetic neuron (wheel speed plus Gaussian noise at signal-to-noise ratios 0.1–4) replacing one of 10 random neurons of a session is recovered at the expected R^2 (dots: mean $\pm$ s.d. over 5 draws and 3 sessions per modality; crosses: the same 10 neurons without it). The decoder reproduces the dorsal-maps GPU decoder (max difference 2×10^{-15}) and the scikit-learn reference (1×10^{-7}). The wheel-speed target agrees with brainbox's filtered speed ($r \geq$0.9999) once encoder glitches are removed. **b** Timing at full resolution: correlation of neural activity with wheel speed at lags -1 to 1 s (positive: neural activity follows the wheel), mean over the 5 % most wheel-correlated neurons selected on the first half of each session and measured on the second half; 2 sessions per modality. Mesoscope deconvolved activity peaks 0.15–0.25 s after wheel speed, BWM spikes 0.10 s before it. **c** What the R^2 measures: predicting each bin from its mean over the training trials (no neural data) gives R^2 0.46 (mesoscope sessions, median) and 0.44 (BWM), far above decoding from all neurons of a bin (current decoder; one dot per session). Most of the variance is the stereotyped time course after movement onset; trial-to-trial differences are 53% (mesoscope) and 56% (BWM) of the variance. **d** Decoder variants, 10 random neurons per 0.4 mm bin (mean $\pm$ s.e.m. over sessions): current (leave-one-row-out choice of the ridge penalty), leave-one-trial-out (LOTO) choice, neural activity of the previous, same and next bin as features (lags), trial-specific speed (bin means of the training trials removed) without and with lags, and the trial-specific target with trials shuffled (null). Mesoscope 0.015 $\rightarrow$ 0.026 with lags, BWM 0.144 $\rightarrow$ 0.218. **e** Same from all neurons of a bin: mesoscope 0.096 $\rightarrow$ 0.132, BWM 0.249 $\rightarrow$ 0.324; trial-specific speed 0.041 vs 0.085. **f** Gain from the lagged features per (session, bin), all neurons.

Discussion

Where the two datasets can be compared they agree, and where they differ the differences follow from what each method samples. Secondary observations:

- **Animals, sessions and regions.** Which mouse and which session are recorded explain more of the variance in decoding than which region, and session-to-session variation is large even within one mouse. Regions differ within a mouse, but the ranking of regions is partly mouse-specific (region $\times$ mouse 11–17% of the variance), so the region effect shared across mice is small. This may reflect differences in imaging fields within a region, in labelling, or in individual strategy. For region maps it means that pooling animals that each contribute few regions mixes region with animal-specific differences; balanced designs, or mixed models with animal, session and region $\times$ animal terms, are needed to compare regions.
- **Stimulus-side tuning.** The mesoscope has about twice the fraction of stimulus-side-tuned neurons of superficial BWM (15% vs 7.5%; Fig. S3). Part of it is one mouse's SSp-ul hotspot, and given how much decoding varies between animals (Fig. 3), this difference should be taken with care.
- **Frontal motor cortex.** MOp and MOs have the highest median scores for choice, feedback and wheel speed in both BWM versions, and mesoscope feedback decoding is strongest frontally (Figs. 2, 3). This agrees with the BWM decoding paper.
- **Map agreement.** Bin by bin, mesoscope and depth-limited BWM maps correlate positively for all five targets (r_s 0.05–0.32), but none is significant against the spatial null (Fig. S4). With only 25 shared bins this test is underpowered; it does not show that the maps disagree.
- **Timing.** Mesoscope activity follows wheel speed by 0.15–0.25 s, as expected from calcium kinetics, while BWM spikes lead it by about 0.1 s; previous- and next-bin features help both (Fig. 7).
- **Cell sampling.** Imaging favours L2/3 somata and includes many low-signal ROIs, while spike sorting keeps all well-isolated units, including fast-spiking interneurons, which often follow movement. Matching cell classes (e.g. by waveform for BWM, by ROI quality for the mesoscope) would test how much of the wheel-neuron difference this explains.
- **Quality control.** Wheel-encoder overflow in 16 mesoscope sessions (fixed) and task–imaging clock offsets in 2 (excluded) were found along the way and should be fixed at the source.

References

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Methods

Data.

- **Task.** The IBL biased visual decision-making task [1]: mice turn a wheel to move a grating of varying contrast to the centre of a screen, and the prior probability of left vs right stimuli switches in blocks.
- **Mesoscope:** IBL two-photon mesoscope sessions of dorsal cortex, accessed from Alyx with ONE [3]: ROI positions (`mpciROIs.mlapdv`), ROI types and activity (`mpci.ROIActivity`) with frame and per-plane times. 151 of the 165 canonical sessions, from 12 mice (Supplementary notes).

- **BWM:** the 2023 data release [2], loaded with the `brainwidemap` package [4] (`load_good_units`, `load_trials_and_mask`) 351 insertions in 288 sessions, good units only. Brain regions are Allen CCF labels mapped with `iblatlas` [5].
- **Quality control.** 16 of 153 mesoscope sessions had wheel-encoder overflow jumps ($\pm$6.6 million rad), which were corrected; no BWM session was affected.

Analysis windows. The wheel-speed bins are identical for both datasets. BWM stimulus, choice and feedback windows follow the BWM single-cell analyses (0.1, 0.1 and 0.2 s). Mesoscope frames are about 0.2 s apart, so its windows are widened to 0.3 s to always contain at least one frame (Fig. S1).

Neurons, trials and spatial bins.

- **Mesoscope neurons:** are ROIs classified as neurons, labelled isocortex (Allen Cosmos level) and with a valid position. Activity is the deconvolved trace.
- **BWM units:** good units labelled isocortex; activity is the firing rate (spike count / window length).
- **Excluded mesoscope sessions:** two sessions whose task and imaging clocks disagree. In both, the cross-correlation of population activity with wheel speed peaks at -146 s and -338 s instead of near 0, and every variable decodes at chance. Trials outside the imaging period are dropped (19 trials of one further session).
- **Trials:** BWM criteria for both datasets: all event times present, a choice was made, and reaction time between 0.08 and 2 s. Sessions need at least 50 such trials (median 347 for the mesoscope and 414 for BWM).
- **Spatial bins:** each neuron is assigned to one hexagonal bin by its ML–AP Allen CCF position relative to bregma, 1 mm across. For decoding, both hemispheres are folded onto the right one.
- **Depth:** depth below the pial surface is the Allen top surface at the unit's ML–AP position minus its DV. Histology-registered mesoscope sessions image 174–318 μm deep (5–95%). The depth-limited BWM set keeps units within 640 μm.

Features and targets.

- One value per neuron and trial: its mean activity in the window. For the mesoscope, each neuron's trace is linearly interpolated at its own sampling times, which include the per-plane time offset, before averaging.
- **Stimulus side:** left vs right, including 0% contrast trials. **Choice:** left vs right wheel turn. **Feedback:** reward vs error.
- **Wheel speed:** mean absolute wheel velocity in each of six 0.2 s bins, paired with the activity in the same bin. All bins of all trials are decoded together.

Decoders and cross-validation.

- A separate decoder is fit for every session $\times$ bin, using either all neurons of that session in the bin or a random draw of 20 neurons (bins with at least 20 neurons, mean over 10 draws).
- **Outer loop:** 5-fold cross-validation over trials, with the same folds for every bin of a session. For binary targets the folds are stratified by class. Features are z-scored using training trials only.
- **Binary targets:** L2-regularised logistic regression with balanced class weights. The regularisation strength (6 values, C from 10^{-4} to 10) is chosen by an inner 3-fold cross-validation on the training trials, maximising balanced accuracy. The score is the balanced accuracy of the pooled held-out predictions (chance 0.5).
- **Wheel speed:** ridge regression; the penalty (7 values, 0.1 to 10^5) is chosen by exact leave-one-trial-out error (rows of a trial are correlated), and the activity of the previous, same and next 0.2 s bin are the features for each wheel bin. Folds keep all bins of a trial together. The score is the R^2 of the pooled held-out predictions (chance 0). Trial-specific wheel speed subtracts the per-bin mean speed of the training trials from the target.
- The decoders run on the GPU and reproduce scikit-learn's `LogisticRegressionCV` and `RidgeCV` pipelines. For sets with more neurons than trials, the logistic decoders are fit in the row space of the trials, which is exact.

Averaging.

- **Decoding maps:** the value of a bin is the mean score over all sessions that contribute at least 20 neurons to it.
- **Scatter plots:** the same per-bin means, for the bins covered by both datasets.
- **Headline numbers:** means over all session $\times$ bin pairs.
- **Tuning maps:** neurons are pooled across sessions within a bin; the value is the fraction with $p < 0.05$, shown for bins with at least 10 neurons.

Single-neuron tuning test.

- The BWM single-cell test [4] (`single_cell_stats`, re-implemented on the GPU; Supplementary notes), applied to the same windows. Mann–Whitney U statistics compare the two conditions within strata (block $\times$ choice for stimulus side, block $\times$ stimulus side for choice and feedback), summed and normalised.
- The null distribution comes from 3,000 permutations of the condition labels within consecutive blocks, and p is the rank of the observed statistic. Zero-contrast trials do not enter this test.
- Sessions without biased blocks (55 mesoscope sessions) are tested within their single 0.5 block. Neurons that cannot be tested are left out of the fractions.

Region and session variance (Fig. 3).

- **Mesoscope scores:** one all-neuron score per session and Beryl region (all neurons of the session in the region, at least 20), with the decoders of Fig. 2.
- **BWM per probe:** one score per probe from all its isocortex units (at least 20), assigned to the Beryl region 300 μm below its brain entry point (histology-aligned trajectory). Angled probes drift, so this is the majority region of the probe's superficial units for only about two thirds of probes.
- **BWM decoding paper:** the published Brain Wide Map decoding results [2,6] (stage-2 tables), one score per session and region from all units of the region, with no minimum unit count (median 16). Each score comes with a p-value against 100–200 pseudo-sessions and the median of that null, which is the baseline in Fig. 3. These tables have wheel velocity instead of trial-specific wheel speed.
- **Regions:** the 15 with at least 3 mesoscope sessions and 3 BWM probes.
- **Shuffle control (mesoscope, BWM per probe):** the same decoders with the trials of all labels and of wheel speed permuted jointly, neural data untouched; 1 permutation per mesoscope score and 3 per probe, pooled per region. This plain trial permutation is less strict than the pseudo-session null of the decoding paper, which keeps the block structure, so the band can sit slightly lower, mainly for stimulus side and choice. For wheel speed the shuffled R^2 is not zero: the stereotyped speed profile after movement onset is still predicted from the stereotyped neural time course.
- **Statistics, mesoscope only.** In BWM nearly every session has one probe, so session variance cannot be separated from probe-to-probe variance. The decoding-paper tables do not help either: few sessions have scores in two or more of these regions, and those regions are usually adjacent along one probe. All-neuron scores grow with the number of neurons, so log neuron count is a covariate in the mixed model and is regressed out before the permutation tests.
- **Per mouse:** the same model without the animal term, score $\sim$ log N + (1|region) + (1|session), fitted separately in each mouse with at least 4 sessions (11 of 12 mice). To separate mouse-specific from shared region effects, the pooled model was also fitted with an additional region $\times$ mouse term (reported in the text, not in Fig. 3b).
- **Session within mouse (permutation):** as the session test below, but scores are centred per region and mouse, and session labels are shuffled within region and mouse, so differences between mice do not contribute.
- **Mixed model:** score $\sim$ log N + (1|region) + (1|animal) + (1|session), crossed random effects, REML (statsmodels [7]). Each component's share is its variance over the total, and each is tested by a likelihood-ratio test against the model without it ($\chi^2_{0/1}$ mixture). Uncertainty in (session + animal share) $-$ region share comes from an animal-level bootstrap: 200 resamples of animals with replacement (a repeated animal and its sessions count as new ones), each refitted with a GPU implementation of the same REML model (`reml.py`, matches statsmodels to 4 decimals).

- **Permutation tests:** the same question without a statistical model, using only reshuffling, on the scores with log N regressed out.
  - *Region:* each session's mean is subtracted from its region scores, so a good session or animal cannot make its regions look good. The statistic is η^2, the share of the remaining variance that lies between regions. The null comes from 5,000 shuffles of the region labels within each session (sessions with at least 2 regions).
  - *Session:* the mirror image. Each region's mean is subtracted, η^2 is the share between sessions, and session labels are shuffled within each region.
  - η^2 is above 0 even for random labels, and more so with many groups (about 150 sessions vs 15 regions). The figure therefore shows adjusted η^2 = ($\eta^2 -$ null mean) / (1 $-$ null mean): 0 is chance level and positive values are the share beyond chance. p is the fraction of shuffles with η^2 at least as large as observed.

Supplementary material

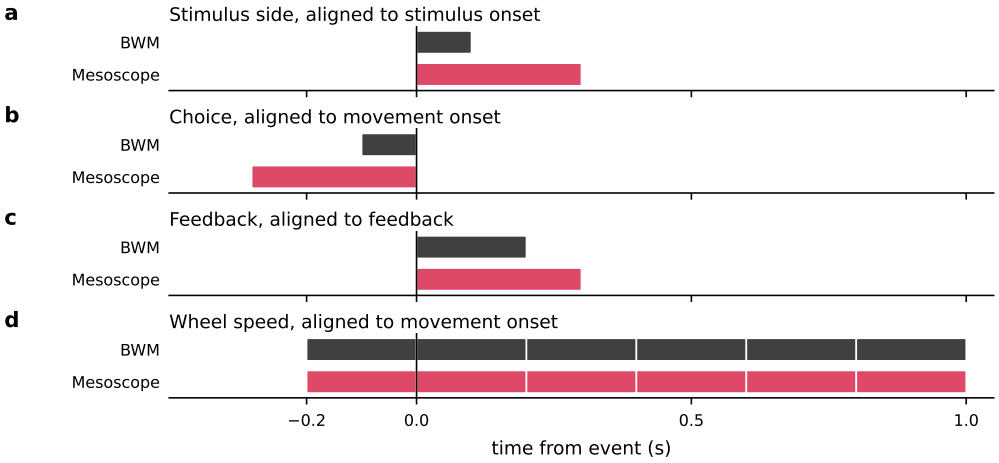


Figure S1: Analysis windows used for decoding and for the single-neuron tuning test. BWM windows follow the Brain Wide Map single-cell analyses. Mesoscope frames are about 0.2 s apart, so the mesoscope stimulus, choice and feedback windows are widened to 0.3 s to always contain at least one frame; the mesoscope value is the mean of the deconvolved trace, linearly interpolated at each neuron's own sampling times, over the window. BWM values are firing rates (spike count / window length). **a** Stimulus side: activity in window $[0, 0.1]$ s (BWM) or $[0, 0.3]$ s (mesoscope) relative to stimulus onset. **b** Choice: window $[-0.1, 0]$ s (BWM) or $[-0.3, 0]$ s (mesoscope) relative to movement onset. **c** Feedback: window $[0, 0.2]$ s (BWM) or $[0, 0.3]$ s (mesoscope) relative to feedback. **d** Wheel speed: six 0.2 s bins from -0.2 to 1 s around movement onset, identical for both modalities; each bin's activity is paired with the mean absolute wheel velocity in the same bin.

Coverage at 1 mm. The depth-limited BWM row is sparse: most bins hold one or two sessions (Fig. S2).

Fraction of tuned neurons. The mesoscope has about twice as many stimulus-side-tuned neurons as BWM, whether or not BWM is restricted to superficial units (Fig. S3).

Bin-wise agreement. Top: mean scores per row. Bottom: per-bin mesoscope against depth-limited BWM, for the 25 bins covered by both (20 neurons), with spatial-null p-values. Wheel speed and feedback agree best (r_s 0.32 and 0.30), but nothing survives the spatial null ($p_{\text{sp}} \geq 0.12$; Fig. S4).

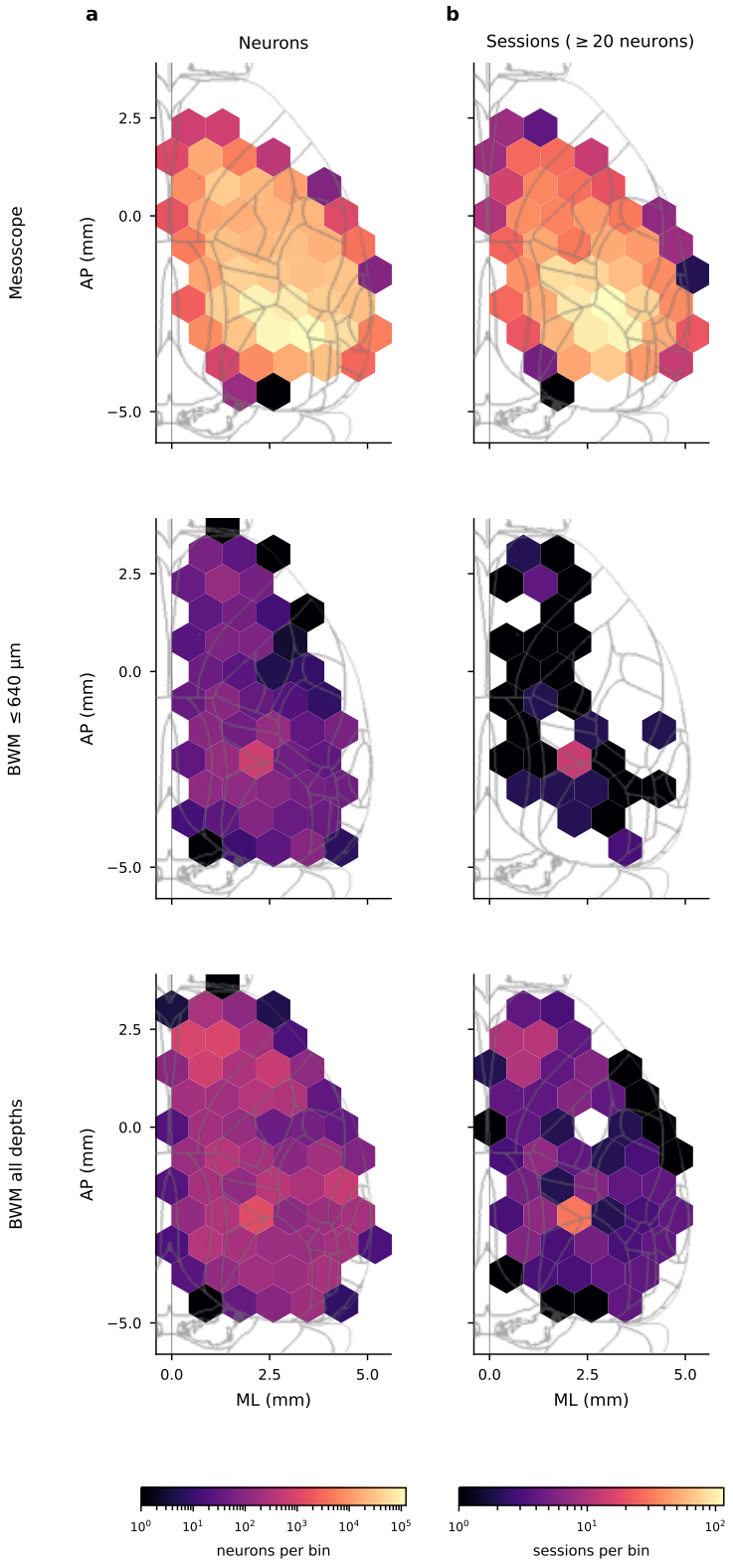


Figure S2: Coverage: mesoscope two-photon imaging (top row; deconvolved activity of isocortex neurons) vs Brain Wide Map Neuropixels units within 640 μm of the pial surface (middle row; twice the depth range the mesoscope images, 174–318 μm) and at all depths (bottom row; reference). Each neuron is assigned to a hexagonal bin (1 mm vertex to vertex) by its ML and AP Allen CCF position; hemispheres folded onto the right. Grey lines: Allen top-view region boundaries. Colour scales are shared across rows. **a** Number of isocortex neurons per bin: mesoscope 1,071,534 neurons, 151 sessions; BWM $\leq$640 μm 2,929 neurons, 204 sessions; BWM all depths 13,002 neurons, 288 sessions. **b** Number of sessions contributing at least 20 neurons to each bin.

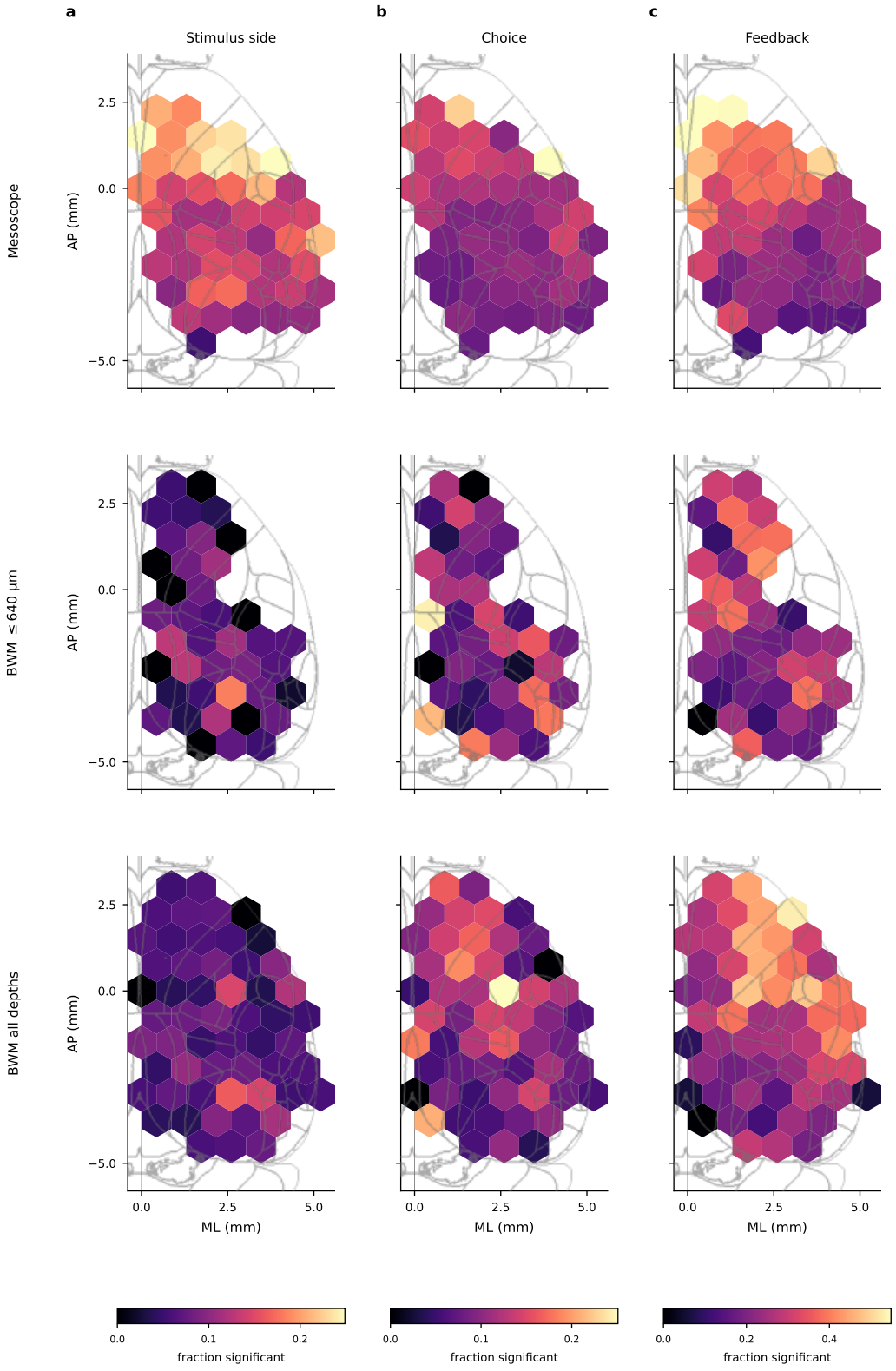


Figure S3: Tuning results: mesoscope two-photon imaging (top row; deconvolved activity of isocortex neurons) vs Brain Wide Map Neuropixels units within 640 μm of the pial surface (middle row; twice the depth range the mesoscope images, 174–318 μm) and at all depths (bottom row; reference). Each neuron is assigned to a hexagonal bin (1 mm vertex to vertex) by its ML and AP Allen CCF position; hemispheres folded onto the right. Grey lines: Allen top-view region boundaries. Colour scales are shared across rows. Single-neuron tuning: the BWM single-cell test (brainwidemap single_cell_stats, re-implemented on the GPU and checked against the original), applied identically to both modalities to the mean activity in each window. Mann–Whitney U statistics are computed within condition strata, summed and normalised; the null distribution comes from 3000 permutations of the condition labels within consecutive blocks, and p is the rank of the observed statistic. Mesoscope windows are 0.3 s long because the frame period is about 0.2 s. **a** Fraction of neurons per bin tuned to stimulus side ($p < 0.05$; bins with $\geq$10 neurons pooled across sessions); window $[0, 0.1]$ s (BWM) or $[0, 0.3]$ s (mesoscope) relative to stimulus onset. **b** Choice; window $[-0.1, 0]$ s (BWM) or $[-0.3, 0]$ s (mesoscope) relative to movement onset. **c** Feedback; window $[0, 0.2]$ s (BWM) or $[0, 0.3]$ s (mesoscope) relative to feedback.

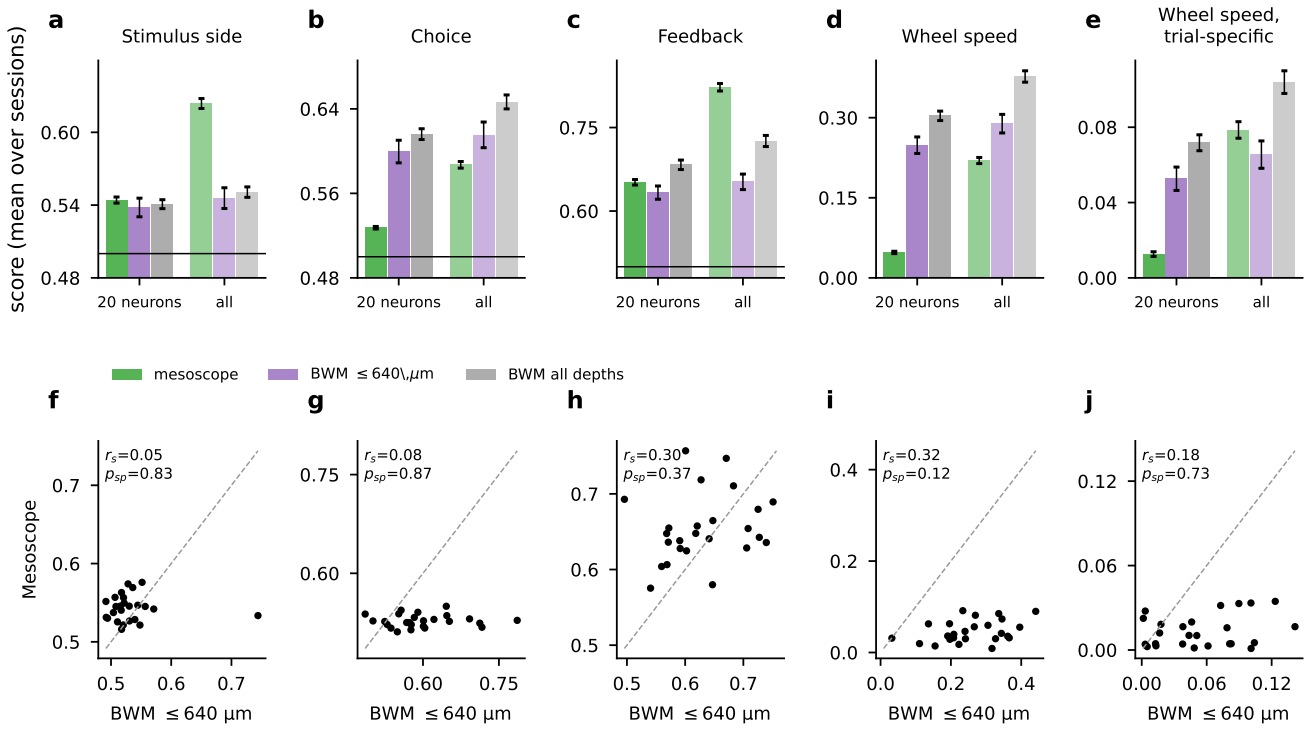


Figure S4: Summary of the revised decoding. **a–e** Mean over sessions ($\pm$ s.e.m.) of each session's mean score over its 1 mm bins, from 20 random neurons (left, solid) and all neurons (right, light). **f–j** Per 1 mm bin, 20 neurons: BWM $\leq$640 μm (x) vs mesoscope (y), the 25 bins covered by both; Spearman r_s, naive p, and p_{sp} against 1000 variogram-matched surrogates of either map (larger p reported). Decoding per session and bin (sessions with $\geq$50 trials and $\geq$20 neurons in the bin), 5-fold cross-validation over trials. Binary variables: L2 logistic regression, balanced accuracy (chance 0.5). Wheel speed in six 0.2 s bins from -0.2 to 1 s around movement onset: ridge regression on the activity of the previous, same and next bin (penalty by exact leave-one-trial-out error), cross-validated R^2; trial-specific: the same with the mean speed profile of the training trials subtracted, so only trial-to-trial differences in speed count. **a** Stimulus side, 20 neurons: mesoscope 0.544, BWM $\leq$640 μm 0.538, BWM all depths 0.541; all neurons: mesoscope 0.624, BWM $\leq$640 μm 0.546, BWM all depths 0.551. **b** Choice, 20 neurons: mesoscope 0.527, BWM $\leq$640 μm 0.600, BWM all depths 0.616; all neurons: mesoscope 0.587, BWM $\leq$640 μm 0.615, BWM all depths 0.647. **c** Feedback, 20 neurons: mesoscope 0.652, BWM $\leq$640 μm 0.633, BWM all depths 0.683; all neurons: mesoscope 0.822, BWM $\leq$640 μm 0.652, BWM all depths 0.726. **d** Wheel speed, 20 neurons: mesoscope 0.048, BWM $\leq$640 μm 0.248, BWM all depths 0.303; all neurons: mesoscope 0.220, BWM $\leq$640 μm 0.289, BWM all depths 0.377. **e** Wheel speed, trial-specific, 20 neurons: mesoscope 0.013, BWM $\leq$640 μm 0.053, BWM all depths 0.072; all neurons: mesoscope 0.079, BWM $\leq$640 μm 0.065, BWM all depths 0.104. **f** Stimulus side: r_s = 0.05, naive p = 0.80, p_{sp} = 0.83. **g** Choice: r_s = 0.08, naive p = 0.69, p_{sp} = 0.87. **h** Feedback: r_s = 0.30, naive p = 0.15, p_{sp} = 0.37. **i** Wheel speed: r_s = 0.32, naive p = 0.11, p_{sp} = 0.12. **j** Wheel speed, trial-specific: r_s = 0.18, naive p = 0.38, p_{sp} = 0.73.

Supplementary notes

- **Tuning p-values.** `mpciROIs.taskTunedP` is not on Alyx and the BWM single-cell p-values are not in the public aggregates, so both modalities use a GPU re-implementation of the BWM single-cell test (`brainwidemap single_cell_stats`). On a 60-neuron check it agreed with the original code on $p < 0.05$ for 97–100% of neurons. The official p-values can be swapped in if someone has the files.
- **Mesoscope region labels.** These are the latest labels on Alyx. Only 25 of 165 sessions have histology-registered labels; the rest use the estimate.
- **Sessions.** 151 of 165 canonical mesoscope sessions. Nine (SP072, SP075, SP076) have no region or position data on Alyx, three have timestamp mismatches, and two have task–imaging clock offsets. BWM: 2023 release, 351 insertions in 288 sessions, good units only.
- **Code.** `int-brain-lab/mesoscope`, `pilot_analyses/dorsal_maps` (branch `dorsal-maps`), with a README listing the pipeline and choices; this report is built by `build_report_v7.py`.